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# Bioavailable phosphorus across Florida's diverse soil orders: implications for crop productivity and environmental protection

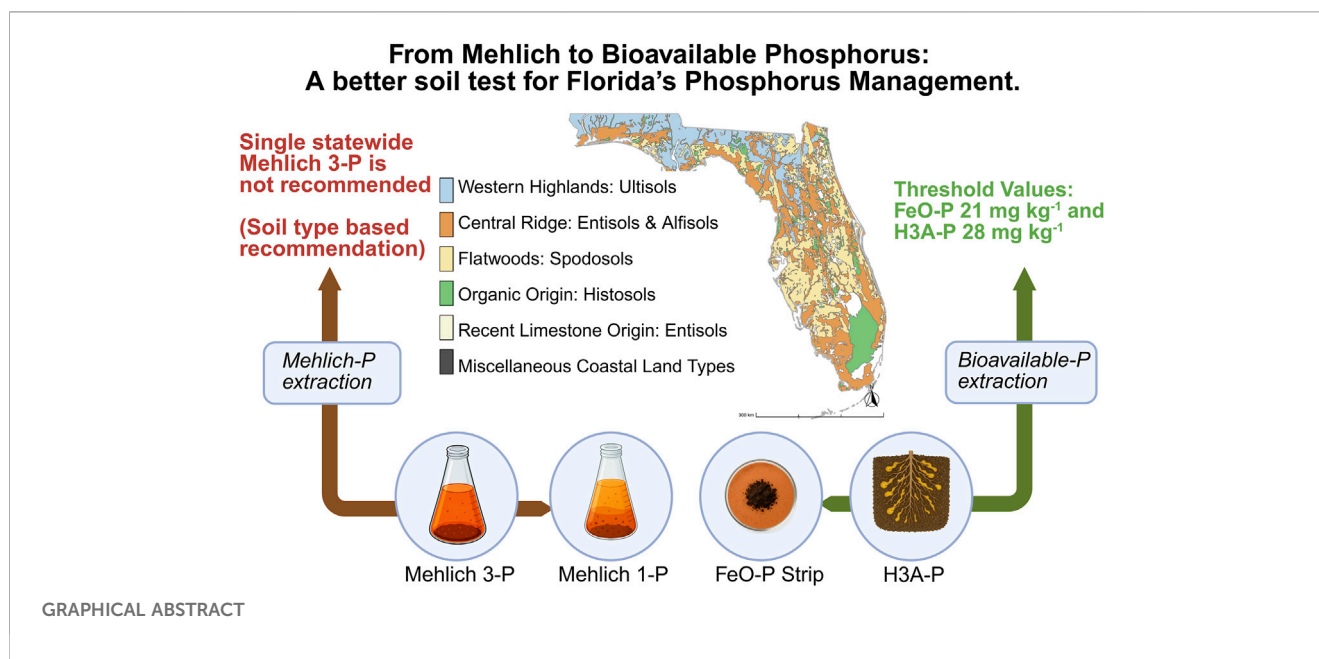
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Effective phosphorus (P) management is critical for sustainable agricultural production in Florida. The state's diverse acid-mineral soils, including Alfisols, Entisols, Spodosols, and Ultisols, exhibit distinct physiochemical properties that profoundly influence P dynamics. This variability can lead to imbalanced P fertilizer applications, which can impair crop nutrient uptake and elevate the risk of environmental P loss. To address this challenge, a comprehensive statewide study of 1,711 surface soils (0–15 cm) was conducted across 14 regions in Florida. Our study evaluated the applicability of established agronomic thresholds for bioavailable P indicators, including the iron oxide strip method (FeO-P), the Haney test (H3A-P), and the conventional Mehlich-1 and Mehlich-3 extractions. Results revealed that the Mehlich 1-P and Mehlich 3-P relationship varied significantly by soil order; the Mehlich 3-P concentration corresponding to the 30 mg kg<sup>-1</sup> Mehlich 1-P benchmark ranged from 41.8 mg kg<sup>-1</sup> in Spodosols to 105.2 mg kg<sup>-1</sup> in Ultisols. Mehlich-3 concentrations determined by ICP-OES were consistently higher than those measured using colorimetric methods, regardless of the soil order. The results confirm that a single, statewide agronomic Mehlich 3-P threshold is inadequate, as it could lead to overapplication in low P-fixing soils or deficiency in high P-fixing soils. Instead, P recommendations must be tailored to regional soil characteristics to ensure optimal crop productivity and minimize P loss to adjacent water bodies. Statewide bioavailable P thresholds are FeO-P = 21 mg kg<sup>-1</sup> and H3A-P = 28 mg kg<sup>-1</sup>, both independent of the soil order. Integrating these thresholds with the Soil Phosphorus Storage Capacity framework provides a dual strategy to enhance crop nutrition while reducing environmental P loss.

## KEYWORDS

fertilizer recommendation, Haney H3A-P, iron oxide strip P, Mehlich 1-P, Mehlich 3-P, soil order



## 1 Introduction

In the United States (US), sandy soils are extensively present in Florida, Nebraska, Michigan, Texas, Georgia, Wisconsin, and Minnesota (Bockheim et al., 2020). Sandy soils are defined by a high proportion of sand particles, typically more than 68% sand and less than 18% clay in 100 cm of the solum (Hartmann and Chinabut, 2005). In Florida, most surface soils are classified as sandy, loamy sand, or sandy loam in texture (Scalera et al., 2019). Sandy soils are characterized by low nutrients and water retention, making them prone to drought and wind erosion. As a result, effective management practices are essential for successful crop production (Yost and Hartemink, 2019). With the application of scientifically tailored fertilizer and nutrient management strategies, Florida sandy soils can achieve high agricultural productivity and support sustainable cropping systems suited to local soil and climatic conditions (Liu et al., 2015).

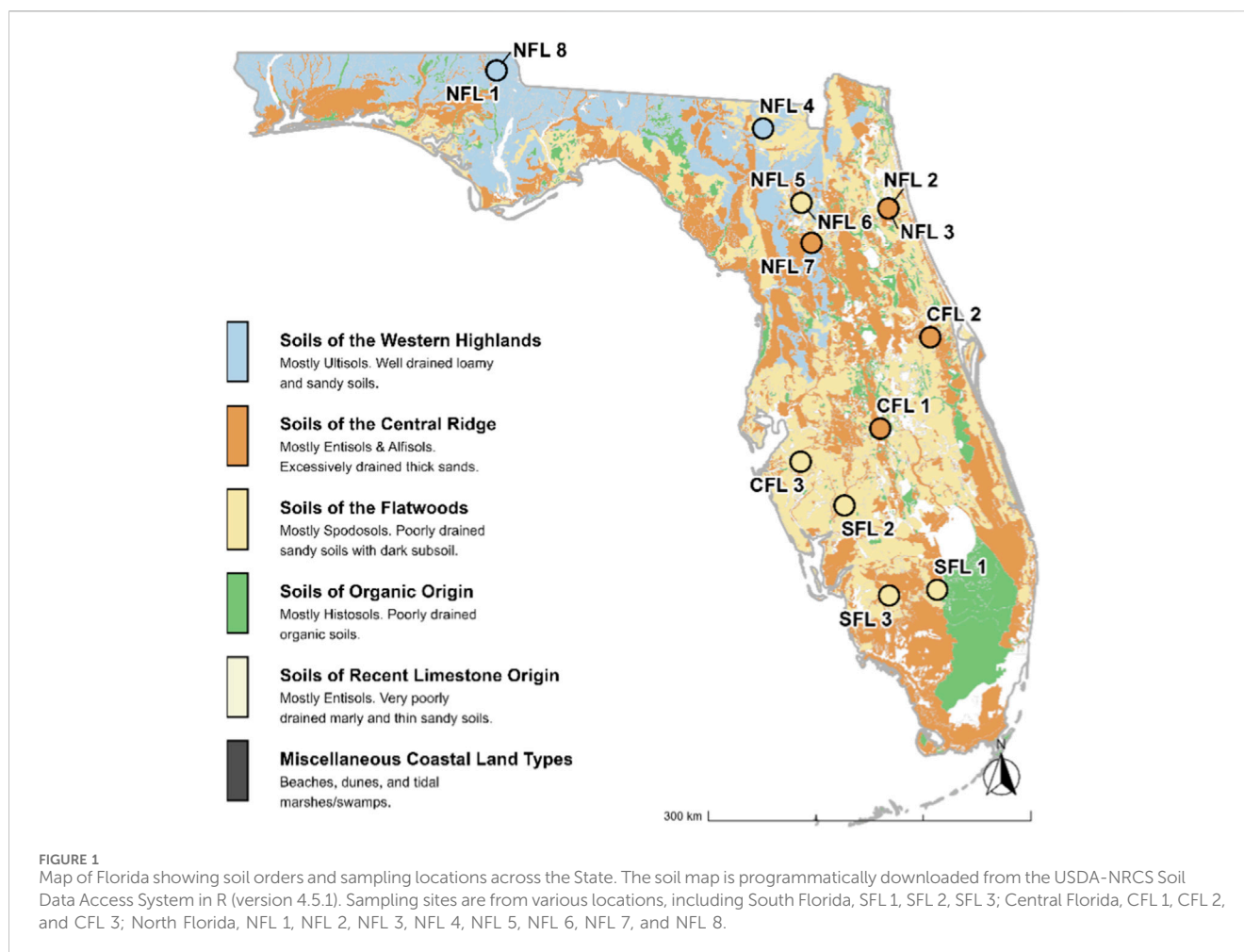
Since the early 2000s, phosphorus (P) applications in the US have steadily increased. For instance, the Economic Research Service (ERS) estimated that US phosphate fertilizer use in corn (*Zea mays*) production has shown a notable upward trend, rising by about 28% between 2000 and 2018 (Economic Research Service U.S. Department of Agriculture, 2025). Over-application of P is particularly acute in Florida. Repeated annual P applications, especially in high-value cropping systems such as potatoes (*Solanum tuberosum*), have resulted in a significant accumulation of legacy P. This has resulted in M3-P levels as high as 475 mg kg<sup>-1</sup>, far exceeding traditional agronomic thresholds (Van Zeghbroeck et al., 2021).

Phosphorus fertilization is essential for maximizing crop yield and supporting vital plant physiological processes (Khan et al., 2023). However, excessive P applications can lead to substantial accumulation of legacy P. Long-term monitoring in subtropical regions has demonstrated that legacy P accumulation can persist for decades (Withers et al., 2018). This surplus P frequently exceeds

crop uptake and through runoff and/or leaching, contributes to environmental degradation including eutrophication, harmful algal blooms, and losses in aquatic biodiversity (EPA. U.S. Environmental Protection Agency, 2025). Inefficient use of P results in higher agricultural input and elevates costs for pollution mitigation (Sena et al., 2020; Sarkar et al., 2024). To address these problems, in addition to management strategies that involve adjusting soil chemical properties and conservation practices, reliable soil tests to correct soil P deficiency are necessary (McDowell et al., 2024; Phuyal and Nair, 2025). Phosphorus needs to be applied only when necessary. Therefore, accurate soil assay becomes a crucial factor in determining sustainable crop production.

The North American Proficiency Testing program utilizes various P testing methods at university laboratories in the southeastern US. The testing program includes Mehlich 1 (M1), Mehlich 3 (M3), Bray P1, Bray P2, and Modified Morgan for P soil analysis (Sera, 2005). The agronomic threshold at which P fertilizer is required is when the M1P value is less than 30 mg kg<sup>-1</sup> (Nair et al., 2004). Florida transitioned to M3 from M1 in 2010 as the State's official soil extractant. The current University of Florida Institute of Food and Agricultural Sciences (UF/IFAS) recommendation is based on a categorical index. For instance, a medium soil P level is between (15–30) mg kg<sup>-1</sup> of M1-P, which is equivalent to (30–47) mg kg<sup>-1</sup> of M3-P. The conversion of M1-P recommendation to M3-P was based on a linear correlation (Mylavarapu et al., 2014).

The current UF/IFAS M3-P recommendation provides a useful framework that could benefit from further refinement to address its limitations. For instance, traditional soil test correlation models often fail to accurately predict crop yield responses to the amount of P applied (Reed et al., 2025). Instead of a linear relationship between M1-P and M3-P, change point analysis can be useful for determining the breakpoint value, beyond which the degree of relationship between variables changes (Morales et al., 2023; Rodriguez et al., 2024; Filippi et al., 2025). Although widely used, the M3-P method is



not considered a reliable approach for estimating bioavailable P, as it extracts both organic P compounds and orthophosphate from soils. This inclusion of organic P introduces variability by overestimation of plant-available P, which can affect measurement accuracy (Cade-Menun et al., 2018). The M3-P extracted pool does not represent the immediately soluble P fraction that poses the primary environmental risk (Zheng et al., 2015). Soil acidification levels strongly influence the P sorption and availability determined by the M3 method (Szara et al., 2018). Additionally, discrepancies in P values have been observed when comparing spectroscopic analysis to inductively coupled plasma optical emission spectroscopy (ICP-OES). Given the analytical and methodological limitations of M3 P testing, an alternative P assessment method is warranted for use across all soil orders in Florida's sandy soils.

A bioavailable form of P analysis may provide a more accurate assessment of soil P levels relevant to actual crop needs. One method to analyze the bioavailable P indicator is the iron oxide impregnated paper strip method (FeO-P) test. This test is beneficial because it gives an index of soil P that plants can access, and it is relatively insensitive to variations in soil type and fertilizer source compared to other conventional assays (Schryer et al., 2024). FeO-P is effective for both acidic and calcareous soils, and shows strong correlation with Olsen-P, Bray 1, and resin-P extraction methods for determining plant-available P (Indiati et al., 2002). FeO-P showed a strong

predictive capability for crop yield and exhibited significant correlations with established soil P extraction methods, including M1-P, M3-P, and Olsen P (Menon et al., 1996). Other tests, such as using Olsen-P, P levels increase as soil pH decreases, suggesting that these tests may overestimate available P at low pH (Pedersen et al., 2023). Additionally, Olsen-P is highly susceptible to colorimetric interference from dissolved organic matter making this method not suitable for agricultural soils (Kowalenko and Babuin, 2007).

Although FeO-P is considered the gold standard for plant-available P, it is rarely used in commercial laboratories due to time and financial constraints (Rodriguez et al., 2024). As an alternative, the Haney soil health (H3A-P) test utilizes malic, citric, and oxalic acids to simulate the organic acids produced by plant roots. Considering the enormous economic costs, time-consuming analysis, and highly correlated data, it has been suggested that H3A-P can replace FeO-P (Haney et al., 2016). The H3A method also allows comparison with conventional M3 extractants, as both are useful for extracting several nutrients. Studies found that M3 extracted greater amounts of calcium (Ca), magnesium (Mg) (Rutter and Ruiz Diaz, 2020), potassium (K) (Rogers et al., 2019; Rutter and Ruiz Diaz, 2020), and higher but highly correlated P (Rutter and Ruiz Diaz, 2020; Mattila and Rajala, 2021) compared to H3A extractants. However, the H3A-P method offers distinct advantages over M3-P: it is insensitive to soil pH (Crittenden et al., 2024; Haney et al., 2006), and because it mimics root exudates, it

allows the simultaneous extraction of inorganic nitrogen (N), which M3 cannot (Haney et al., 2017; Rogers et al., 2019). Despite this huge potential, there is limited information available on P recommendations based on these bioavailable P methods.

Florida does not have just one soil order per region (north, central, south); rather, seven soil orders are distributed throughout the state (Figure 1). Spodosols are the most extensive (27%–34%) in the state, followed by Entisols (23% of the state) and Ultisols (20% of the state). Histosols are organic-rich soils present around the Everglades Agricultural Area, while Alfisols, Inceptisols, and Mollisols occupy smaller areas (Nunes, 2025). Given Florida's unique geochemical conditions, developing locally calibrated soil tests requires an understanding of soil morphology. Soil morphology determines P retention and release dynamics, which in turn determine P availability to the crop and potential loss from the soil.

Soil Phosphorus Storage Capacity (SPSC) is a quantitative measure used to estimate how much additional P a soil can absorb before reaching a threshold where further P additions pose a significant risk of water pollution through runoff or leaching (Equation 1) (Nair and Harris, 2004). The SPSC is a more comprehensive risk indicator because, unlike soil test P or phosphorus saturation ratio (PSR) (Nair, 2014), it can identify the risk of P loss even in soils that have a naturally low P-holding ability but have not been previously impacted. This makes SPSC a true measure of P risk for any soil group, as long as its specific PSR threshold is known (Nair and Harris, 2014). The surface soil across Florida has a threshold PSR of 0.10, with a 95% confidence interval of 0.05–0.15. It marks the point where soil changes from being a “P sink” to a “P source” (Dari et al., 2018).

$$\text{SPSC} = (\text{Threshold PSR} - \text{Soil PSR}) \times \left( \frac{M3 \text{ Fe}}{56} + \frac{M3 \text{ Al}}{27} \right) \times 31 \quad (1)$$

Integrating PSR and SPSC into P management decisions is crucial for optimizing P use while minimizing environmental risks.

The overarching goal of this research is to compare relationships among M1-P, M3-P, FeO-P, and H3A-P and to evaluate the reliability of these extraction methods in Florida sandy soils. The specific objectives are to 1) to develop and refine M3-P thresholds for major Florida soil orders: Alfisols, Entisols, Spodosols, and Ultisols, 2) obtain M3-P values as determined by ICP-OES using a benchmark M1-P value of 30 mg kg<sup>-1</sup> across diverse soil orders, 3) compare M3P determined by ICP-OES with colorimetric methods, 4) establish statewide thresholds for FeO-P and H3A-P as alternative indicators of bioavailable P that enable statistically reliable comparisons with conventional Mehlich metrics, and 5) measure the risk of P loss across Florida soils using the SPSC framework to support environmentally responsible nutrient management practices.

## 2 Materials and methods

### 2.1 Analytical methods

Phosphorus in soil samples was quantified using several extraction methods:

For M1, the procedure was performed as described (Mehlich, 1953). Five g of air-dried soil, sieved to less than 2 mm, was extracted with 20 mL of M1-P extraction solution (0.05M HCl + 0.0125M H<sub>2</sub>SO<sub>4</sub>), shaken for 5 min at low speed 180 cpm (cycles per min (cpm)) at room temperature. The slurry was immediately filtered through a medium-porosity filter paper (Whatman™ Qualitative Filter Paper: Grade 2). Phosphorus concentration in the extracted samples was analyzed using ICP-OES, with blanks and standards prepared in the M1 extracting solution.

For M3, analysis was performed following Mehlich (1984) method. Two g of air-dried, ground soil is extracted with 20 mL of M3 extracting solution (0.2M CH<sub>3</sub>COOH, 0.015M NH<sub>4</sub>F, 0.013M HNO<sub>3</sub>, 0.001M EDTA, and 0.25M NH<sub>4</sub>NO<sub>3</sub>), shaken for 5 min (120 cpm). The slurry was filtered through a medium-porosity filter paper Whatman #41. The extracted P samples along with Iron (Fe) and Aluminum (Al) were analyzed by ICP using a blank and standards prepared in the M3 extracting solution. The extracted samples were analyzed colorimetrically at 880 nm using a spectrophotometer (EasyPlus UV/VIS, Mettler Toledo GmbH, Greifensee, Switzerland) with blanks and standards prepared in the Mehlich-3 extracting solution.

For H3A-P, the procedure from Haney et al. (2017) was used. Two g of air-dried, ground soil was extracted with 20 mL of H3A-4 extracting solution (0.0024 M C<sub>6</sub>H<sub>8</sub>O<sub>7</sub>·H<sub>2</sub>O, 0.004 M C<sub>4</sub>H<sub>6</sub>O<sub>5</sub>, 0.004 M C<sub>2</sub>H<sub>2</sub>O<sub>4</sub>·2H<sub>2</sub>O), shaken for 10 min (120 cpm). The slurry was centrifuged, and the resultant suspensions were centrifuged at 3500 revolutions per minute (rpm) for 5 min. After that, the extracts were filtered through Whatman 2V filter paper (8-micron filter paper) and analyzed for P by ICP using a blank and standards prepared in the H3A-P extracting solution.

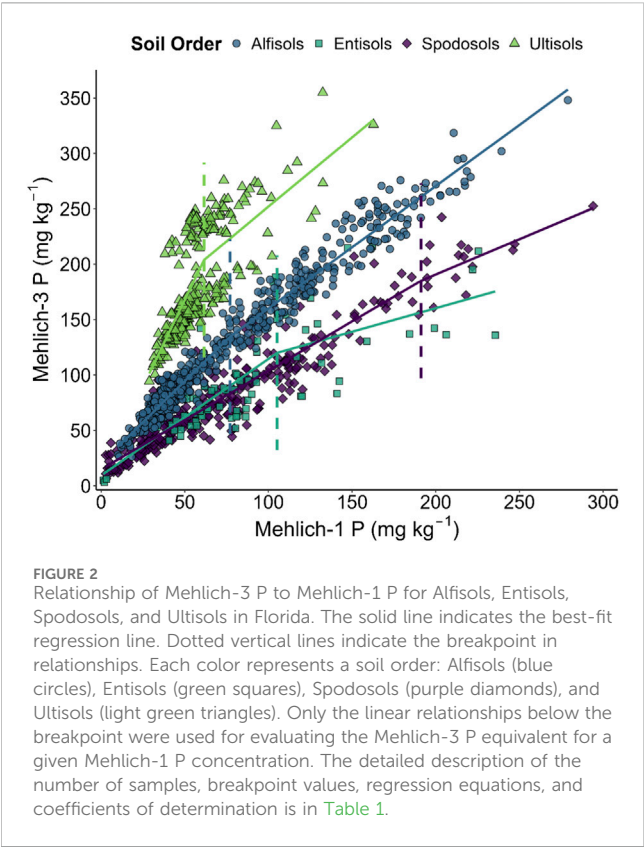
For water-soluble P (WSP), 2 g of air-dried, ground soil was extracted with 20 mL of double-deionized water (DDI), shaken for 60 min (180 cpm), centrifuged the suspensions at 4,000 rpm for 10 min, and filtered the extracts through a 0.45 μm filter paper. The samples were analyzed for P by colorimetry using a blank and standards prepared in the DDI extracting solution, similar to the M3 colorimetric method.

For FeO-P, 1 g of soil was placed into a graduated glass jar with FeO-P-impregnated filter paper secured between screens. FeO-P-impregnated filter papers were prepared by soaking Whatman no. 50 filters in 0.65 M ferric chloride solution overnight. After an initial drying period, the filters were dipped in 2.7 M NH<sub>4</sub>OH to precipitate the iron oxide, rinsed in DDI water, and allowed to dry completely. Eighty mL of 0.01 M CaCl<sub>2</sub> was added to the jar containing the soil, and it was shaken in an orbital shaker for 16 h at 265 rpm. The filter papers were then removed, dried, and extracted in 50 mL of 0.1M H<sub>2</sub>SO<sub>4</sub>, shaken in a shaker at 265 rpm for 1 h. The samples were analyzed by colorimetry using a blank and standards prepared in the 0.1 M H<sub>2</sub>SO<sub>4</sub> extracting solution. Following desorption, the P in the extract was quantified using the Murphy and Riley (1962) colorimetric method.

### 2.2 Soil sampling

A total of 1,711 soil samples (0–15) cm were collected between 2023 and 2024 from major agricultural regions across Florida. Sampling sites are from various regions including South Florida, SFL 1 (n = 39), SFL 2 (n = 40), SFL 3 (n = 149), Central Florida, CFL 1 (n = 80), CFL 2 (n = 6) and CFL 3 (n = 92); North Florida, NFL 1

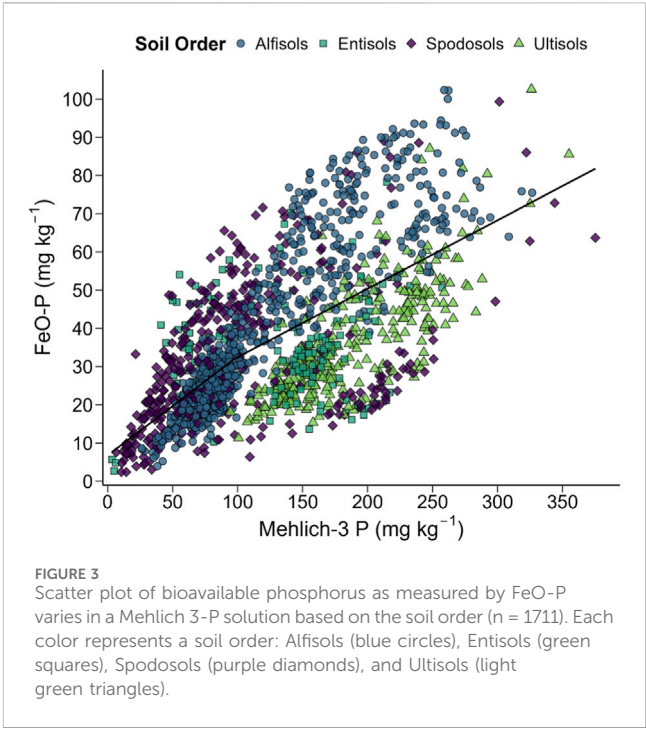




(n = 251), NFL 2 (n = 375), NFL 3 (n = 440), NFL 4 (n = 32), NFL 5 (n = 16), NFL 6 (n = 72), NFL 7 (n = 71), and NFL 8 (n = 48) (Figure 1). The collected soil represents four major soil orders (Alfisols, Entisols, Spodosols, and Ultisols) and various cropping systems, including potatoes, tomatoes (*Solanum lycopersicum*), and snap beans (*Phaseolus vulgaris*) across Florida. All the collected samples were air-dried at 25 °C and then sieved through a 2 mm mesh prior to analysis.

2.3 Statistical analysis

Statistical analysis and data visualization were conducted using R (version 4.5.1) (R Core Team, 2025), and figures were generated with the ggplot2 package (Wickham, 2016). A segmented regression



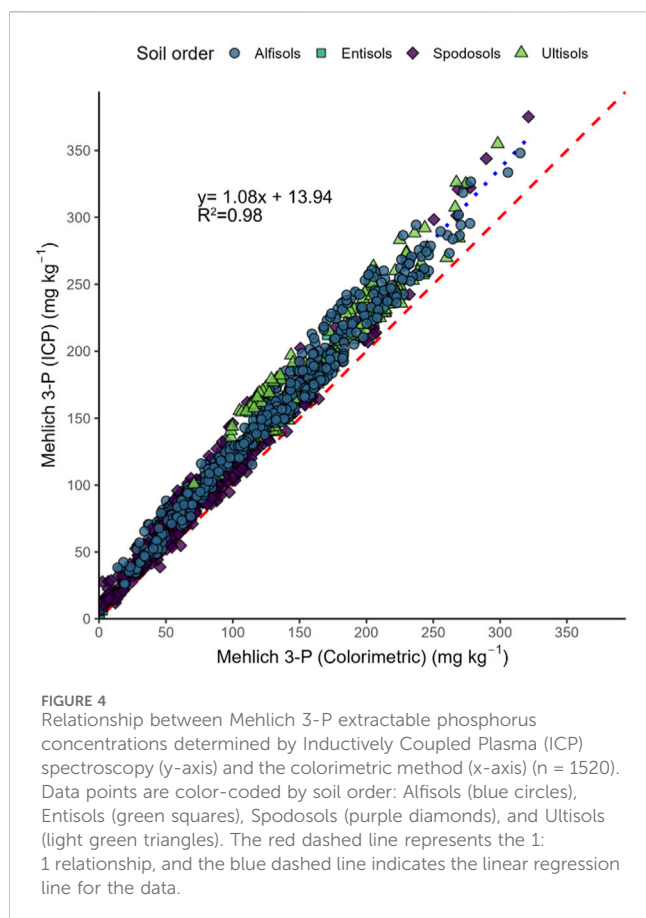
model (also known as a breakpoint or change-point model) was applied following previous literature (Rodriguez et al., 2024). Soil samples were grouped, and models were fit independently for the four soil orders: Alfisols, Entisols, Spodosols, and Ultisols. This nonlinear model was used to identify a critical threshold (breakpoint) where the relationship in slope changes. Model diagnostics were conducted by examining Pearson residuals for nonlinearity and heteroscedasticity using the ggResidpanel package. The model was fitted using the Levenberg-Marquardt algorithm for nonlinear least squares via the nlsLM function from the minpack.lm package (Elzhov et al., 2023). After model fitting, 95% confidence intervals (CIs) were estimated for all model parameters and breakpoints. Since the UF/IFAS P fertilizer threshold is based on the M1-P value being below 30 mg kg<sup>-1</sup>, corresponding values for M3-P were determined. The linear relationship below the breakpoint in the segmented regression model was used to do the calculations.

To investigate the relationship between Florida soils and the bioavailable P extraction methods (FeO-P and H3A-P), a simple

TABLE 1 Common soil orders, the breakpoint in the relationships between Mehlich 1-P and Mehlich 3-P, and the equation before the breakpoint from which Mehlich 3-P corresponds to a given Mehlich 1-P. The breakpoint is the Mehlich 1-P value (mg kg<sup>-1</sup>) at which the slope of the Mehlich-3 versus Mehlich-1 relationship changes significantly.

| Soil order | Obs. (n) | Breakpoint (mg kg <sup>-1</sup> ) | Equation before the breakpoint | R <sup>2</sup> value for the equation | Recommended M3-P value (mg kg <sup>-1</sup> ) |
|------------|----------|-----------------------------------|--------------------------------|---------------------------------------|-----------------------------------------------|
| Alfisols   | 815      | 77.0                              | y = 1.48x + 20.44 ***          | 0.83                                  | 64.8                                          |
| Entisols   | 86       | 105.3                             | y = 1.02x + 10.96 ***          | 0.51                                  | 41.7                                          |
| Spodosols  | 331      | 191.2                             | y = 0.88x + 15.37 ***          | 0.84                                  | 41.8                                          |
| Ultisols   | 283      | 61.5                              | y = 3.12x + 11.62 ***          | 0.42                                  | 105.2                                         |

In the table, Significance levels: \* = P < 0.1; \*\* = P < 0.01; \*\*\* = P < 0.001.



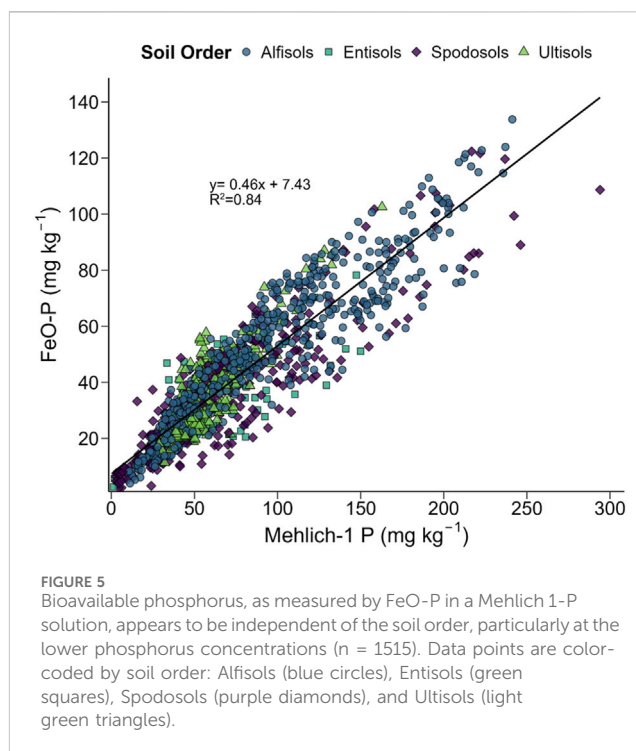
linear regression analysis was conducted. The corresponding value of bioavailable P extraction methods when the M1-P (a previously threshold soil test P) was  $30 \text{ mg kg}^{-1}$ , was determined. Similarly, the relationship between M3-P extractable P concentrations, as determined by ICP-OES and colorimetric methods, was compared using simple linear regression analysis. Model assumptions included the linearity of the relationship, the normality of residuals, and homoscedasticity. Model significance was tested using an F-test, and slope significance using a t-test, with a  $p$ -value  $< 0.05$  considered statistically significant.

To compare the mean concentration of extractants (moles of Al and Fe) between soil orders, and the mean values of each extraction method (M1-P, M3-P, H3A-P, and FeO-P) across the different soil orders, Analysis of variance (ANOVA) was conducted. When significant effects were detected, Tukey's Honestly Significant Difference (HSD) *post hoc* test was implemented to identify pairwise differences between soil orders, with a  $p$ -value  $< 0.05$  considered statistically significant.

### 3 Results and discussion

#### 3.1 Relationship between Mehlich 1-P and Mehlich 3-P

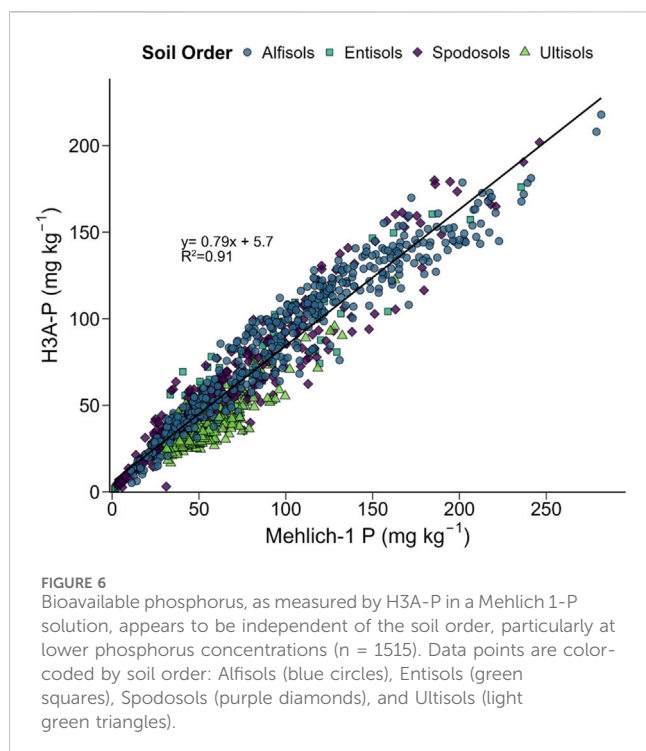
The relationship between M1-P and M3-P varies across soil orders (Figure 2; Table 1). For instance, when the M3-P



concentration reaches  $105.2 \text{ mg kg}^{-1}$  in Ultisols, P fertilization is required. In contrast, Spodosols require fertilizer application at a much lower M3-P level of  $41.8 \text{ mg kg}^{-1}$ , which is approximately 60% lower than the critical value observed for Ultisols. Overall, P availability in the M3-P extract at a given M3-P concentration follows the trend Spodosols = Entisols < Alfisols < Ultisols, which aligns with the M3-P recommendations derived from M3-P calibration (Table 1).

The different P fertilizer application thresholds for different soil orders are primarily due to variations in the presence of P-binding minerals. In the present study, the presence of Al (also the moles of combined Al and Fe) in the soil followed the order: Ultisols > Alfisols > Entisols = Spodosols, and the presence of Fe followed the order: Alfisols > Entisols = Ultisols > Spodosols (Supplementary Tables S2, S4, S6). It appears that the influence of Al is stronger than that of Fe in retaining P in soil. A previous study found that Al is more responsible for P retention in sandy soil compared to Fe (Borggaard et al., 1990). Another study by Kadir et al. (2022) also concluded that Al is often the dominant factor controlling P retention in podzolic soils.

Commercial laboratories frequently opt for a single extractant across different soil types to streamline the analytical process. This approach not only reduces operational costs but also enhances efficiency when processing large volumes of soil samples originating from diverse geographic region (Hochmuth et al., 2014; Zhang et al., 2021). For Florida, the M3-P threshold recommendation was initially estimated using a linear relationship with M1-P (Mylavarapu et al., 2014). However, several studies reported that the correlations between M1 and M3 extracted P are not perfect (Sera, 2005; Sandhu et al., 2023; Rodriguez et al., 2024). This relationship can vary widely depending on soil properties such as soil type, pH, and mineralogy, and simple conversions may not perform consistently across different soils. This



variability underscores that a single, universal M3-P recommendation is not suitable, and soil-specific calibration is necessary for accurate interpretation. The results (Figure 2; Table 1) suggest a need for soil order-specific P recommendations.

## 3.2 Relationship between Mehlich 3-P and FeO-P

The bioavailable P measured as FeO-P in M3-P extract also differs among soil orders, further supporting the challenge of establishing a uniform M3-P threshold across soils. The M3-P extractant determines all P forms extracted by the M3 solution and tends to overestimate plant-available P in the soil (Figure 3), which is attributed to its strong acid composition that dissolves insoluble P forms. The efficiency of M3-P extraction largely depends on soil mineralogy, Fe and Al oxide content, and pH, which control the solubility and binding strength of phosphate ions (Szara et al., 2018). For example, Ultisols are highly weathered soils that can have enormous P sorption capacities compared to Entisols (Yang and Post, 2011). Consequently, M3-P often extracts more P from Ultisols than from Entisols, even when their bioavailable P is similar. These limitations underscore that M3-P and FeO-P target different P pools.

### 3.2.1 Analytical discrepancies of spectroscopic analysis to inductively coupled plasma optical emission spectroscopy

M3-P concentrations determined by ICP-OES were consistently higher than those obtained via colorimetric methods, irrespective of the soil order ( $P < 0.001$ ) (Figure 4). This discrepancy increases at higher P concentrations.

At the lower P concentrations (below the threshold P value), M3-P determines primarily inorganic P, while at the higher P

concentrations, M3-P extracts all P forms, including organic-bound P (Cade-Menun et al., 2018). Furthermore, in M3-P, ICP determines the total P in the solution. In contrast, colorimetric methods (e.g., the molybdate blue technique) are interpreted to measure only orthophosphate in solution (Cade-Menun et al., 2018; Ziadi et al., 2025). This discrepancy could have resulted in a different amount of P extraction between the two analytical methods, particularly at higher P concentrations.

In the southern states of the US, both methods are common; however, the ICP method is replacing the colorimetric method. Still, many of the original fertilizer recommendations were made from the colorimetric method (Sikora, 2014). Throughout this paper, M3-P values are reported based on ICP analysis, as it is the most used technique in analytical laboratories. Therefore, accurate interpretation of M3-P data requires that both the extraction conditions and the method of analysis be reported, as each can affect the magnitude of the values obtained.

## 3.3 Bioavailable phosphorus extraction in Florida sandy soils

Bioavailable P, as determined using the FeO-P procedure, shows less variation across soil orders in M1-P extract. The breakpoint in the M1-P and M3-P relationship was determined statistically, and the linear relationship below this breakpoint was used to estimate the FeO-P value corresponding to the previously threshold M1-P value of 30 mg kg<sup>-1</sup>. Based on this relationship, M1-P of 30 mg kg<sup>-1</sup> corresponds to a FeO-P of 21.12 mg kg<sup>-1</sup> (95% CI = 20.49–21.75; n = 1515,  $P < 0.001$ ) (Figure 5). For practical applications, a FeO-P value of 21 mg kg<sup>-1</sup> can be used as a threshold for crop P requirements, where soils with FeO-P < 21 mg kg<sup>-1</sup> require fertilizer application, while values >21 mg kg<sup>-1</sup> indicate a higher risk of P loss.

Similar to FeO-P, P determined using the H3A-P procedure also showed a smaller variation compared to the M3-P extractant (Figure 6). The statistically derived relationship between M1-P and H3A-P shows that a M1-P value of 30 mg kg<sup>-1</sup> corresponds to a H3A-P value of 27.95 mg kg<sup>-1</sup> (95% CI = 27.12–28.69; n = 1515,  $P < 0.001$ ). For practical use, an H3A-P threshold of 30 mg kg<sup>-1</sup> can serve as a guideline for P management, where soils with H3A-P < 28 mg kg<sup>-1</sup> require P fertilizer application, while those with H3A-P > 28 mg kg<sup>-1</sup> indicate an increased risk of P loss.

Florida's diverse sandy soils require soil-type-insensitive methods to accurately measure P for statewide recommendations. This is not possible with the traditional M3-P as it is highly sensitive to soil type (pH, organic matter, variability in Al and Fe content, etc.) as discussed above. M1-P seems to be a better soil test particularly for the agronomic P determination, as its relationship with bioavailable P shows a stronger linear relationship (Figures 5, 6). Since the bioavailable P forms are insensitive to soil characteristics, the optimum rate can be calculated.

Contemporary research emphasizes the need to adopt soil-type-insensitive P methods. Studies on Orthic Black Chernozem and Gleyed Black Chernozem soils have shown that H3A-P can be recommended to a broad range of soil properties (soil pH, soil organic matter content, clay content, and calcium carbonate equivalent). The authors compared H3A-P with Olsen-P and determined a recommendation rate, where H3A-P values below

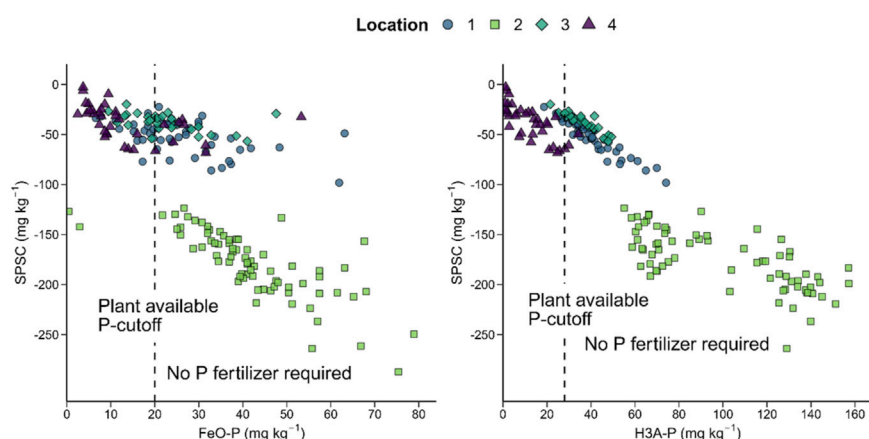


FIGURE 7

The relationship between agronomic indicators of bioavailable phosphorus (H3A-P-right, FeO-P left) and SPSC, an environmental phosphorus indicator ( $n = 190$ ). SPSC values are calculated using concentrations of phosphorus, iron, and aluminum extracted via a Mehlich 3 solution, specific to observations within each location. Each color represents a different location: Location 1 (blue circles), Location 2 (light green squares), Location 3 (green diamonds), and Location 4 (deep purple triangles). The vertical dashed lines indicate the plant-available phosphorus cutoff values. Samples to the left of the cutoff represent soils with sufficient phosphorus where no additional phosphorus fertilizer is required. The FeO-P value can be converted to H3A-P and vice versa using the equation  $y = 1.43x + 3.58$ , where  $x$  is the FeO-P value and  $y$  is the H3A-P value.

24 mg kg<sup>-1</sup> do not require P fertilizer (Crittenden et al., 2024). A similar study Jones and Mallarino (2018) suggested keeping soil P levels below H3A-P 16 mg kg<sup>-1</sup> and H3A-P 19 mg kg<sup>-1</sup> for corn and soybean (*Glycine max*), respectively. In a study by Sims et al. (2002) in the Mid-Atlantic region, Ultisols, Entisols, and Inceptisols were found to have FeO-P levels below the 15 mg kg<sup>-1</sup> threshold required for P fertilizer application. More recently, a study found in Florida sandy soils, M1-P 30 mg kg<sup>-1</sup> is equivalent to FeO-P 16.8 mg kg<sup>-1</sup> ( $n = 107$ ) and H3A-P 20.2 mg kg<sup>-1</sup> ( $n = 108$ ) (Rodriguez et al., 2024). The current study updates the previous recommendation by including a larger number of representative samples across Florida sandy soils. Bioavailable P threshold values based on H3A-P and FeO-P can address both: reducing the risk of either excessive fertilization (environmental consequence) or insufficient supply (yield loss).

#### 4 Relating soil phosphorus storage capacity and bioavailable P indicators to predict crop yield

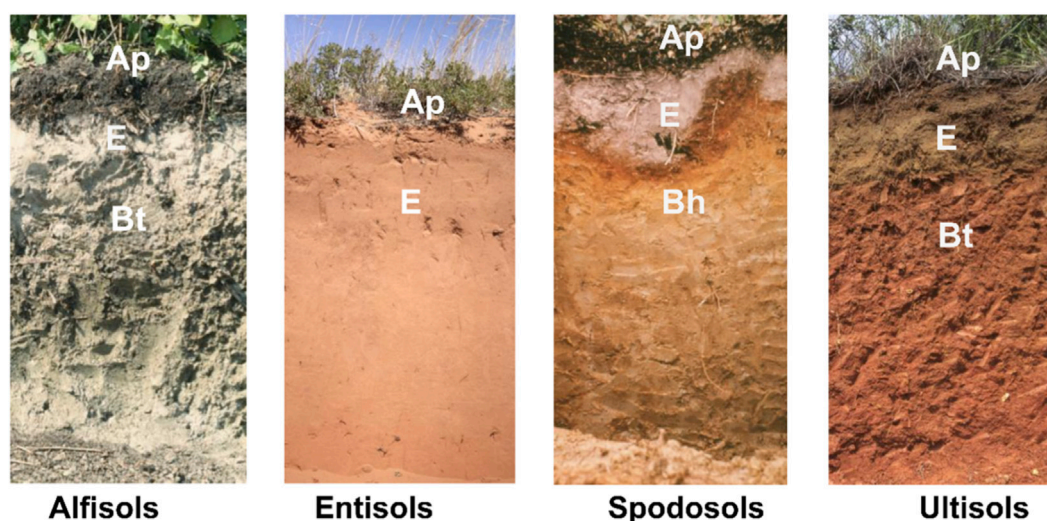
A total of 190 soil samples were collected from four different P-impacted agricultural fields across Florida, with no known history of P application, and analyzed for FeO-P and H3A-P. The SPSC was calculated from P, Fe, and Al in a Mehlich 3 solution using (Equation 1). Figure 7 shows the relationship between SPSC and bioavailable P indicators. All the samples had negative SPSC, and nearly all samples from Location two fell outside the plant-available P threshold for FeO-P and H3A-P. This suggests that P fertilization is unnecessary at Location 2, and P mining from the soil may be a viable option. Using the same statistical approach (Section 2.3), the relationship between bioavailable P indicators and crop yield can be derived. For instance, Phuyal et al. (2025) reported that sweet orange (*Citrus sinensis*) required 26 mg kg<sup>-1</sup>

P based on the FeO-P extraction method. However, achieving the maximum yield corresponded to an SPSC of -34.9 mg kg<sup>-1</sup>. Once P exceeded the recommended P fertilization rate, the SPSC slope became more negative, indicating higher environmental risk.

Soil Phosphorus Storage Capacity is a cumulative parameter that can be quantified over any specified soil depth, provided that accurate measurements of bulk density and sampling depth (Nair et al., 2020). Throughout this manuscript, the calculations are based on surface soils (0–15 cm), which exclude the significant influence of subsurface horizons, such as the Bh horizon in Spodosols, on P movement and retention. Substituting the y-axis in Figure 7 with depth-specific SPSC values enables a more precise assessment of subsurface P contributions to crop yield responses. Therefore, both plant-available P and SPSC metrics can be customized for specific crops and soil conditions to enhance P use efficiency.

While many studies have reported on the impact of inorganic P on crop yield, the long-term environmental implications of organic fertilizers are a distinct and significant concern. Organic amendments such as poultry manure, animal manure, and municipal biosolids are frequently applied to agricultural lands to meet crop nitrogen (N) requirements rather than P-based agronomic rates (Elliott et al., 2002; Chrysostome et al., 2007). Due to the high P content relative to N in these materials, this practice often results in P application far exceeding crop removal (Wei et al., 2022). The repeated application of organic P creates a “legacy P” source (Harris et al., 2010). Study shows heavily manure-impacted dairies continue to be P sources even long after P applications have stopped (Nair et al., 2011). The SPSC predicts crop-available legacy P and serves as an effective indicator of legacy agricultural P (Supplementary Figure S1) (Nair et al., 2020). Therefore, having the SPSC indicator is essential for predicting crop P application.





**FIGURE 8**  
Soil profiles of common soil orders in Florida—Alfisols, Entisols, Spodosols, and Ultisols (with horizon designations superimposed) (USDA-NRCS, 2025b). All these soils are cultivated. Organic soils (Histosols) behave differently and are not included here (Phuyal and Nair, 2025).

## 5 Understanding phosphorus movement through subsurface horizons in Florida sandy soils

Florida soil is predominantly sandy, yet it exhibits substantial heterogeneity in P retention capacity. This variability underscores the necessity for site-specific P management strategies that account for the retention characteristics of the surface and subsurface horizons. Standard agronomic soil testing typically targets the 0–15 cm depth; however, subsurface horizons also differ in their P sorption potential. These differences may influence both crop P uptake and the risk of P loss, particularly when uniform fertilizer P applications are made to surface soils without consideration of subsurface dynamics. Therefore, assessing the profile-specific P retention characteristics of Florida's primary soil orders, as summarized in [Supplementary Table S9](#), is critical for developing effective, site-specific nutrient management strategies.

The nutrient-holding capacity of Florida's Spodosols is generally poorer than that of other soil orders. This is because Florida Spodosols predominantly belong to the aquods suborder, which inherently contains little silicate clay (USDA-NRCS, 2025a). Directly beneath the surface horizon lies an E horizon, which lacks P-retaining minerals (Harris et al., 2010). This structure makes Spodosols prone to leaching even at low P application rates (Dari et al., 2018). The Bh or spodic horizon, however, is dominated by organically-complexed Fe and Al and may act as a P sink (Figure 8) (Chakraborty et al., 2012; Nair et al., 2011). In contrast, Florida Entisols can retain more P due to sand grain coatings in the upper horizons that enhance P retention (Harris et al., 2010). However, these soils exhibit minimal evidence of pedogenic horizon development (Chrysostome et al., 2007), and P can move downward to saturate subsurface layers (Freitas et al., 2025).

Alfisols contain a Bt or Btg horizon with high base saturation (USDA-NRCS, 2015), which contributes to better P retention. Ultisols are characterized by intense weathering and leaching, resulting in a clay-enriched subsoil (USDA-NRCS, 2015). Because clay soil generally holds more P than sandy soils (Machado and Souza, 2012), Ultisols possess the highest P retention capacity, as also observed in the present study.

## 6 Summary and conclusion

This research demonstrates the inadequacy of a single, statewide M3-P threshold for Florida's heterogeneous sandy soils. The reliance on M3-P as a diagnostic metric may present challenges for growers due to regional variability in threshold values. A practical approach would involve recording GPS coordinates for a given site and utilizing a decision-support application to determine the corresponding soil order and associated regional threshold value. The study also revealed the importance of clearly stating the extraction procedure and analytical technique when reporting M3-P, given their substantial influence on the resulting measurements. Analysis of bioavailable P indicators provides a more reliable assessment, leading to the establishment of new statewide agronomic thresholds of 21 mg kg<sup>-1</sup> for FeO-P and 28 mg kg<sup>-1</sup> for H3A-P. When integrated with the SPSC framework, these thresholds support a dual strategy to optimize crop nutrition while reducing environmental P losses. SPSC can be computed for surface and/or subsurface once the concentrations of P, Fe, and Al in a Mehlich-3 extract are obtained from any accredited soil testing laboratory. Future research should incorporate subsurface analyses and apply these validated bioavailable thresholds to develop crop-specific and site-specific P recommendations, thereby advancing sustainable nutrient management.

## Data availability statement

The datasets presented in this article are not readily available because all data generated and analyzed in this study were produced by the authors and collaborators. The datasets supporting the findings are available from the corresponding author upon reasonable request. Requests to access the datasets should be directed to Vimala D. Nair [vdn@ufl.edu](mailto:vdn@ufl.edu).

## Author contributions

VN: Resources, Funding acquisition, Project administration, Writing – original draft, Data curation, Supervision, Investigation, Writing – review and editing, Validation, Conceptualization, Methodology. DP: Software, Writing – review and editing, Formal Analysis, Validation, Writing – original draft, Data curation. LV: Data curation, Validation, Formal Analysis, Methodology, Writing – review and editing, Writing – original draft.

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## Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fenvs.2026.1748291/full#supplementary-material>

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